

Convertible Bond Pricing: Jump-to-Default PDE

One-factor stock PDE with a stock-dependent hazard rate, American conversion, issuer call and holder put – pricing, calibration, numerics, Greeks and acceptance tests.

01 SCOPE

What the engine prices and returns

Price a single convertible bond as one instrument on a one-dimensional grid in the stock price S . Credit enters through a default intensity $\lambda(S, t)$: at default the stock jumps down and the bond pays recovery. Conversion (holder, American), call (issuer, with soft-call trigger and notice), put (holder, Bermudan), discrete coupons and dividends are imposed as constraints or jump conditions on the grid.

Rates, volatility and the hazard curve are deterministic. Stochastic rates, stochastic credit, dilution, resets and exact path-dependent soft calls are out of scope for v1 (Section 09).

OUTPUTS PER VALUATION

Dirty and clean price (points of par)
Parity, conversion premium, bond floor, investment premium
Delta (shares/bond and % of parity), gamma, credit-delta split
Vega, CS01 (total and bucketed), rho, recovery, p , borrow, dividend
Theta / carry, jump-to-default (naked and hedged)

02 MODEL

Dynamics, hazard rate, governing equation

Under the pricing measure the stock follows a diffusion with a single default jump. Before default,

$$\frac{dS_t}{S_{t-}} = (r(t) - q(t) - b(t) + \eta \lambda(S_t, t)) dt + \sigma dW_t - \eta dJ_t,$$

where J jumps from 0 to 1 at default with intensity λ , $\eta \in (0, 1]$ is the fractional stock loss at default ($\eta = 1$: stock goes to zero; default for v1), q the dividend yield and b the stock borrow cost. The $+\eta\lambda$ drift makes the discounted stock (with dividends and borrow) a martingale despite the jump. The hazard rate is stock-dependent:

$$\lambda(S, t) = \min\left(\lambda_0(t) \left(S_0/S\right)^p, \lambda_{\max}\right), \quad p \geq 0.$$

$\lambda_0(t)$ is calibrated to the credit curve at today's spot S_0 (Section 05); p sets how fast credit deteriorates as the stock falls; $\lambda_{\max}$ caps the intensity near $S = 0$ for numerical stability.

GOVERNING PDE (BETWEEN EVENT DATES)

$$\begin{aligned} \frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + (r - q - b + \eta\lambda)S \frac{\partial V}{\partial S} - (r + \lambda)V + \lambda V_D(S, t) &= 0 \\ V_D(S, t) &= \max\left(\kappa(t)(1 - \eta)S, R \cdot N\right) \quad (\text{value received at default}) \end{aligned}$$

Terminal condition: $V(S, T) = \max(\kappa(T)S, N_{\text{red}} + c_{\text{final}})$. Solve backward in time from T to the valuation date t_0 .

- **Discounting at $r + \lambda$** is survival-weighted discounting; the source term λV_D pays back recovery. For a straight bond the effective spread is $\approx \lambda(1 - R)$.
- **Recovery convention** v1 uses recovery of par, $R \cdot N$. Make it a switch: recovery of par plus accrued, or recovery of market value, are term-sheet / desk choices.
- **Units** V is in currency per bond of face N ; S per share; κ = shares per bond. Quote prices as $100 V / N$ points.

03 INPUTS

Term sheet and market data

INPUT	DEFINITION	NOTES
N, N_{red}, T	Face, redemption amount, maturity	Redemption may differ from par (accreting / OID bonds)
Coupons c_i at t_i	Schedule, rate, day count	Accrued $A(t)$ from the same schedule
$\kappa(t)$	Conversion ratio, shares per bond	Time-varying for step-ups or dividend-protection adjustments
Conversion windows	Dates on which conversion is allowed	Default: any time; contingent conversion out of scope
Call schedule	Periods, call price B_c (clean), trigger h , notice n	Hard call: $h = 0$. Soft call: callable if $S \geq h \cdot N / \kappa$
Put schedule	Dates, put price B_p (clean)	Bermudan: only on listed dates
$S_0, q(t)$ or D_k	Spot, dividend yield or discrete cash dividends	Remove protected dividends if κ adjusts for them
$b(t)$	Stock borrow cost	Acts exactly like an extra dividend yield
$r(t)$	Risk-free (funding) curve	Deterministic
CDS / spread curve	Par spreads at pillar maturities	Calibrates $\lambda_0(t)$
$R, p, \eta, \lambda_{\text{max}}$	Recovery, hazard elasticity, jump size, cap	Desk-set parameters; report sensitivities to each
σ	Diffusion volatility of the stock	Calibrated in <i>this</i> model; not a Black-Scholes implied vol

04 EVENTS & CONSTRAINTS

What happens at each grid time

After each backward time step from t_{k+1} to t_k , apply the following in this order. V_{cont} is the PDE solution (continuation value). Call and put prices are dirty: $\tilde{B}_c = B_c + A(t)$, $\tilde{B}_p = B_p + A(t)$. Conversion forfeits accrued unless the term sheet says otherwise.

CONSTRAINT ORDER AT EACH TIME STEP

1. Issuer call (if callable at t and trigger met) $V \leftarrow \min(V_{\text{cont}}, \max(\tilde{B}_c^{\text{eff}}, \kappa S))$
2. Holder put (on put dates) $V \leftarrow \max(V, \tilde{B}_p)$
3. Holder conversion (if convertible at t) $V \leftarrow \max(V, \kappa S)$

With $\tilde{B}_c \geq \tilde{B}_p$ this equals $V = \max(\tilde{B}_p, \kappa S, \min(V_{\text{cont}}, \tilde{B}_c^{\text{eff}}))$: the issuer calls to minimise value, the holder responds by converting if parity exceeds the call price.

- ♦ **Coupons** At a coupon date t_i (stepping backward): $V(S, t_i^-) = V(S, t_i^+) + c_i$. The payment is only received on survival; that is automatic because the grid is the pre-default value.
- ♦ **Discrete dividends** $V(S, t_d^-) = V(S - D_d, t_d^+)$ by cubic interpolation on the grid. If κ adjusts for the dividend, apply the adjustment $\kappa(t_d^-)$ vs. $\kappa(t_d^+)$ instead.
- ♦ **Soft call (Markov approximation)** The 20-of-30-day trigger is path-dependent. v1 treats the bond as callable at t only where $S \geq h \cdot N / \kappa$. Optionally raise h slightly to proxy the averaging requirement. Exact treatment needs an extra state variable (Section 09).
- ♦ **Notice period n** When called, the holder keeps the choice between cash and shares until $t + n$. Replace the call price by the value of that choice (ignoring default during the notice window):
$$\tilde{B}_c^{\text{eff}}(S) = e^{-rn} \tilde{B}_c + \kappa C_{\text{BS}}(S, \tilde{B}_c / \kappa, n).$$
With $n = 0$ this collapses to $\tilde{B}_c$.
- ♦ **Rannacher restart** Every event above is a non-smooth condition. Take two fully implicit (half-size) steps immediately after each one before returning to Crank-Nicolson (Section 06).
- ♦ **Event dates on the grid** Every coupon, dividend, call/put start, and put date must be an exact time-grid node.

Hazard curve, volatility and desk parameters

- ♦ **Hazard curve** $\lambda_0(t)$ Piecewise constant between CDS pillars $T_1 < \dots < T_m$. Bootstrap sequentially: for pillar j , root-find the constant λ_0 on $(T_{j-1}, T_j]$ so the model par spread at S_0 matches the market. Both CDS legs are priced with the same PDE operator ($\kappa = 0$, no constraints):

$$\text{protection } U : \partial_t U + \mathcal{L}U - (r + \lambda)U + \lambda(1 - R) = 0, \quad \text{annuity } W : \partial_t W + \mathcal{L}W - (r + \lambda)W + 1 = 0,$$

with $U(T_j) = W(T_j) = 0$, $\mathcal{L} = \frac{1}{2}\sigma^2 S^2 \partial_{SS} + (r - q - b + \eta\lambda)S \partial_S$, and model par spread $s_j = U(S_0, t_0)/W(S_0, t_0)$. Because $\lambda(S)$ is convex in S , the naive shortcut $\lambda_0 = s/(1 - R)$ overstates longer-dated spreads when $p > 0$; use it only as the root-finder's starting guess.

- ♦ **Volatility** σ Two modes. (a) *Convert-implied*: solve for the σ that reprices the market convert; this is the desk's quoted "implied vol". (b) *Listed options*: calibrate σ by pricing the options in this same jump-to-default model (calls: $V_D = 0$; puts: $V_D = K$ when $\eta = 1$). Never plug in a Black-Scholes implied vol directly: it already contains the default risk carried by λ , so it double-counts.
- ♦ $p, R, \eta, \lambda_{\max}$ Desk inputs. p from a regression of log-spread changes on log-stock changes, a fit to liquid converts, or convention; $R = 40\%$ is a common default. None of these can be hedged cleanly, so each gets a reported sensitivity (Section 07).

Grid, time stepping and early exercise

- ♦ **Space grid** S from 0 to $S_{\max}$, with

$$S_{\max} \geq 4 \max(S_0, N/\kappa) e^{3\sigma\sqrt{T}}.$$

Non-uniform grid (sinh stretching) concentrated around S_0 , the conversion price N/κ and the soft-call trigger. Place S_0 exactly on a node.

- ♦ **Boundary at $S = 0$** The derivative terms vanish; the PDE row reduces to $\partial_t V - (r + \lambda_{\max})V + \lambda_{\max}V_D = 0$ (plus coupons). No extra condition needed.
- ♦ **Boundary at $S_{\max}$** Linearity, $V_{SS} = 0$, with a one-sided first derivative. Do not impose $V = \kappa S$ as a Dirichlet condition: it is wrong when $\kappa = 0$ (bond floor, CDS legs).
- ♦ **Convection** Central differences; switch a node to up-wind where a central-difference coefficient would go negative. The $\eta\lambda S$ drift is large at small S , so this matters there.
- ♦ **Time stepping** Crank-Nicolson with Rannacher start-up (two implicit half-steps at T and after every event). Tridiagonal solve per step. The time grid contains every event date exactly.
- ♦ **Early exercise** Preferred: penalty method (Forsyth-Vetzal), iterating the penalised system to 10^{-8} relative tolerance per step; keeps second-order convergence. Acceptable for v1: explicit projection of the constraints after each step (first-order in time).
- ♦ **Interpolation** Report values at S_0 directly from the node. For other spots, use cubic interpolation, never linear, when computing Greeks.
- ♦ **Convergence target** Halving both ΔS and Δt should move price by < 0.02 points and delta by < 0.001 . Report this check with every model release.

Definitions, bump sizes, conventions

Grid Greeks come from the t_0 solution. Bumped Greeks re-run the full solve on the same space and time grid (to avoid grid noise) with central differences. Any bump to rates, recovery, or the credit curve re-bootstraps $\lambda_0(t)$ with CDS spreads held fixed, unless stated.

GREEK	DEFINITION	METHOD / CONVENTION
Delta	$\Delta = \partial V / \partial S$	Central difference on the grid (non-uniform stencil). Report as shares per bond (Δ) and % of parity (Δ/κ). Hedge = short Δ shares per bond.
Gamma	$\Gamma = \partial^2 V / \partial S^2$	Grid, three-point non-uniform stencil. Also report $\Gamma \cdot S \cdot 1\%$: change in hedge shares per 1% move. Can be negative in the busted zone.

GREEK	DEFINITION	METHOD / CONVENTION
Credit-delta split	Δ_{eq} : delta with λ frozen at $\lambda(S_0, t)$; $\Delta_{\text{cr}} = \Delta - \Delta_{\text{eq}}$	Re-solve with $p = 0$ and $\lambda_0(t) \leftarrow \lambda(S_0, t)$. Shows how much of the hedge comes from the credit channel.
Vega	$\partial V / \partial \sigma$ per 1 vol point	Reprice at $\sigma \pm 1\%$.
CS01	$\partial V / \partial s$ per 1bp	Parallel $\pm 1\text{bp}$ to the CDS curve, re-bootstrap, reprice. Also bucketed by pillar.
Rho / DV01	$\partial V / \partial r$ per 1bp	Parallel $\pm 1\text{bp}$ to the rate curve, re-bootstrap with spreads fixed.
Recovery	$\partial V / \partial R$ per 5 pts	$R \pm 5\%$ with spreads fixed and re-bootstrap: isolates the recovery assumption.
p -sensitivity	Price and delta change	$p \pm 0.25$. Model-risk metric; moves delta materially in the busted zone.
Borrow, dividend	per 10bp	$b \pm 10\text{bp}$, $q \pm 10\text{bp}$.
Theta / carry	1-day roll	Reprice at $t_0 + 1$ business day, market inputs and S unchanged; include any coupon paid. Report carry (coupon accrual $- \Delta \times$ dividends, + short rebate) separately.
Jump-to-default	$\text{JTD} = V_D - V$	Naked loss on default. Hedged: $\Delta \eta S_0 - (V - V_D) + \text{CDS payout} \sum \text{notional} \cdot (1 - R)$.

TRAPS WORTH WRITING INTO THE CODE REVIEW

- **Double-counting default** A Black-Scholes implied vol already prices crash and default risk; using it as σ with $\lambda > 0$ overprices the option and inflates vega.
- **Hazard blow-up near zero** With $\lambda \propto S^{-p}$ the delta can exceed κ at very low S (credit delta dominates; see the reference run below). The size of that effect is set by p and λ_{max} , so treat very-low- S Greeks as model-dependent.
- **Accrued conventions** Clean vs. dirty call and put prices, and whether conversion forfeits accrued, move prices by up to a full coupon. Make them explicit flags, not hard-coded choices.
- **Sticky assumptions** A spot bump holds $\lambda_0(t)$ fixed, so spreads move with S along $\lambda(S)$. That is the intended “credit delta”; document it so traders don’t hedge the same credit move twice.

08 VALIDATION

Analytic checks and a reference run

Common parameters unless stated: $N = 100$, $\kappa = 1$, $S_0 = 100$, $T = 5$, $r = 4\%$, $q = b = 0$, $\sigma = 25\%$, $R = 40\%$, $\eta = 1$, $\lambda_{\text{max}} = 2$. The “model” column is from a prototype of this spec (Crank-Nicolson + Rannacher, projection, uniform $\Delta S = 0.25$, $\Delta t = 1/400$, $S_{\text{max}} = 800$).

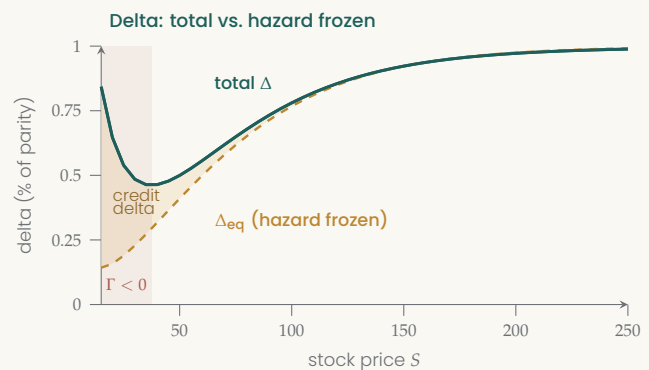
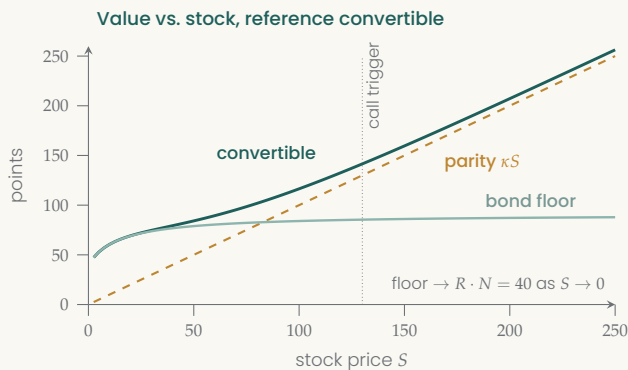
TEST	SETUP AND CLOSED FORM	EXACT	MODEL
T1 No credit	$\lambda_0 = 0$, no coupon/call/put. Early conversion never optimal, so $V = Ne^{-rT} + \kappa C_{\text{BS}}(S_0, N/\kappa, T, r, \sigma)$	112.1857	112.1857
T2 Risky zero	$\kappa = 0$, $p = 0$, $\lambda = 3\%$. $V = Ne^{-(r+\lambda)T} + RN \frac{\lambda}{r+\lambda} (1 - e^{-(r+\lambda)T})$	75.5313	75.5313
T3 Call under JTD	European call, $K = 100$, $p = 0$, $\lambda = 3\%$, $V_D = 0$. Equals Black-Scholes with $r \rightarrow r + \lambda$ (plain BS: 30.3127)	36.9563	36.9563
T4 Limits	Deep in the money: $V \rightarrow \kappa S$, $\Delta \rightarrow \kappa$. $\kappa = 0$: V equals the straight risky bond	qualitative	
T5 Monotonicity	$V \geq \max(\kappa S, \text{floor})$; $\partial V / \partial \lambda_0 \leq 0$; vega ≥ 0 away from the call trigger	qualitative	

Reference convertible. Common parameters plus $\lambda_0 = 3\%$ flat (spread $\approx 180\text{bp}$), $p = 1$, 2% coupon paid semi-annually, callable from year 3 at 100 + accrued when $S \geq 130$, no notice period, no put. A correct implementation should land within about 0.05 points on price and 0.005 on delta. CS01 here bumps λ_0 by $1\text{bp}/(1 - R)$ without re-bootstrapping.

AT SPOT 100	VALUE
Price (pts)	116.44
Parity / premium	100.00 / 16.4%
Bond floor	84.03
Delta (% parity)	0.780
Gamma	0.0044
Vega (per vol pt)	+0.48
CS01 (per bp)	-0.0063
Rho (per bp)	-0.015
JTD naked	-76.4
JTD delta-hedged	+1.6

SPOT	PRICE	DELTA	DELTA EQ	FLOOR
40 · busted	79.38	0.464	0.318	76.81
60	89.44	0.556	0.499	80.48
100 · balanced	116.45	0.780	0.767	84.03
160 · equity-like	168.90	0.937	0.938	86.34

Δ_{eq} freezes λ at its value at that spot. The gap is the credit delta: about 0.15 when busted, zero once equity-like. The bond floor is the same model with $\kappa = 0$, so it also rises with S through $\lambda(S)$.



09 EXTENSIONS

Deliberately left out of v1

- ♦ **Exact soft call** Add a Parisian-style counter of days above the trigger as a second state variable.
- ♦ **Make-whole / takeover** Extra shares from a make-whole table on a change-of-control event; needs an event intensity or scenario overlay.
- ♦ **Contingent conversion, resets** Path-dependent conversion rights and conversion-price resets.
- ♦ **Stochastic rates or credit** Two-factor PDE or Monte Carlo (e.g. Longstaff-Schwartz); matters for long-dated, low-delta bonds.
- ♦ **Dilution and net-share settlement** Firm-value effects of conversion and cash-plus-shares settlement.
- ♦ **Cross-currency** Bond and stock in different currencies adds an FX factor.

ACCEPTANCE CRITERIA FOR v1

- ♦ **Tests** T1-T3 reproduced to 4 decimals; T4-T5 pass across a spot ladder from 10% to 300% of the conversion price.
- ♦ **Reference run** Matches the table above within 0.05 points and 0.005 delta.
- ♦ **Convergence** Grid-halving report attached; price change < 0.02 points.
- ♦ **Risk report** Every Greek in Section 07 produced with its stated bump and convention, plus the credit-delta split and hedged JTD.

Reference numbers come from an independent prototype and are for implementation checking only; model choices (p , recovery convention, soft-call treatment) vary by desk and should be agreed before build.